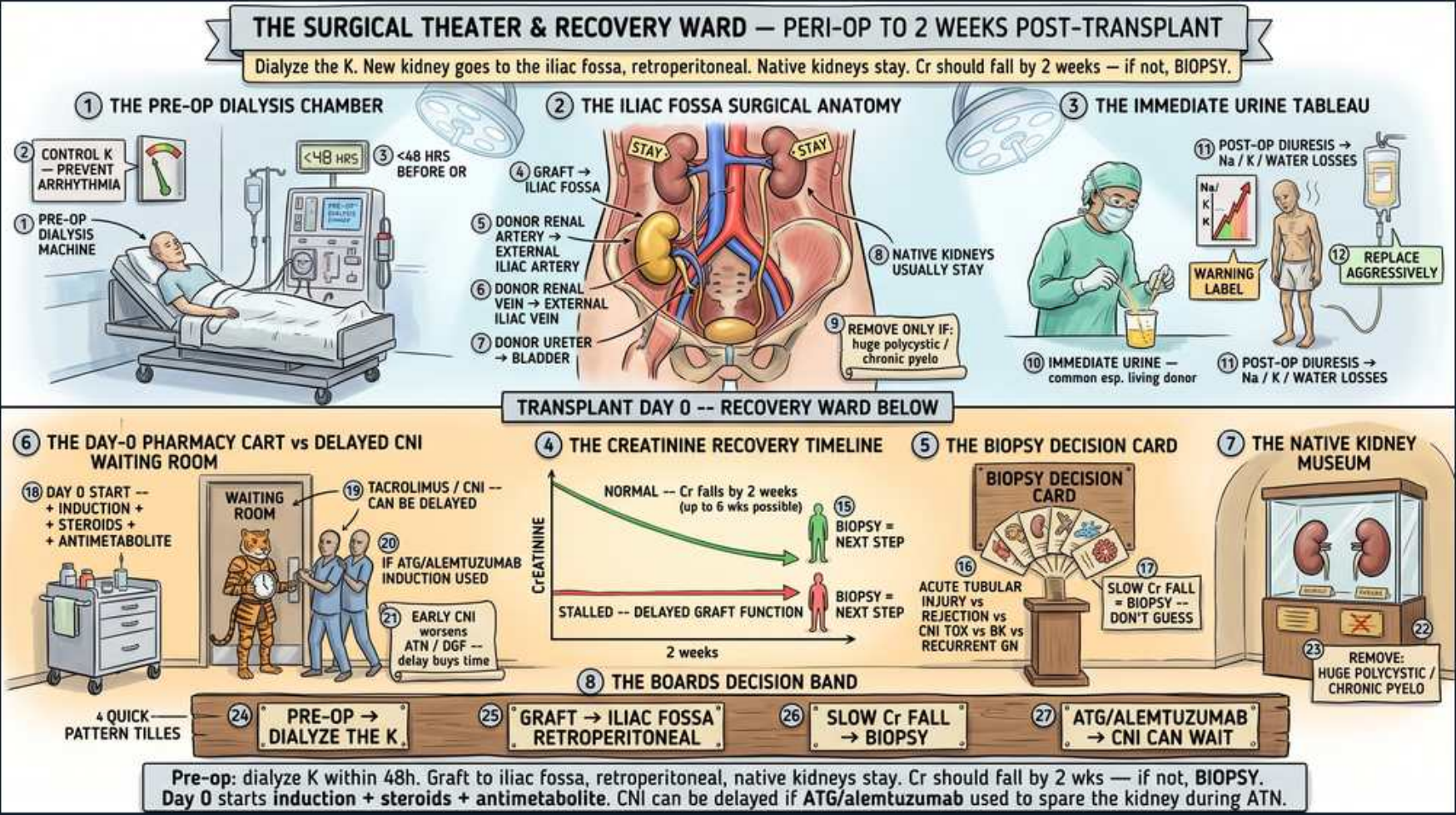


Peri-Op to 2 Weeks Post-Transplant: Surgery, Recovery, Early Graft Function



1. Pre-op: dialyze K if high

WHAT YOU SEE

Pre-op dialysis chamber, control K

WHAT IT ENCODES

Before transplant surgery, correct dangerous hyperkalemia with dialysis (within ~48h pre-op) to prevent intraoperative arrhythmia. Volume and acid-base are optimized. After a working graft, potassium and uremia self-correct as the kidney resumes function.

BOARD TRAP

Skipping pre-op dialysis for a high potassium 'because the new kidney will fix it' — fix dangerous K before surgery to avoid arrhythmia.

2. Graft → iliac fossa

WHAT YOU SEE

Iliac fossa surgical anatomy, graft placement

WHAT IT ENCODES

The new kidney is placed extraperitoneally in the iliac fossa, with the renal artery anastomosed to the external/internal iliac artery, renal vein to the iliac vein, and ureter to the bladder. This superficial location makes the graft easy to biopsy and image.

BOARD TRAP

Expecting the graft in the native retroperitoneal renal bed — it is placed in the iliac fossa (extraperitoneal), which is why it's palpable and easily biopsied.

3. Native kidneys stay

WHAT YOU SEE

Native kidneys usually stay, museum

WHAT IT ENCODES

Native kidneys are usually left in place unless there is a specific indication (refractory hypertension, recurrent infection, large polycystic kidneys, malignancy). The transplant is added, not a swap. Huge polycystic native kidneys are a classic reason to remove.

BOARD TRAP

Assuming native nephrectomy is routine — native kidneys are typically retained; removal is reserved for specific indications like massive PKD or infection.

4. Cr should fall in 2 weeks

WHAT YOU SEE

Creatinine recovery timeline, normal curve

WHAT IT ENCODES

With a well-functioning graft, creatinine should fall to baseline over days to about 2 weeks (faster with living-donor than deceased-donor kidneys). A normal recovery curve is reassuring; a stalled or rising curve prompts evaluation.

BOARD TRAP

Panicking at a slow early Cr drop in a deceased-donor graft — delayed graft function from ATN is common and recovers; the cue is the trend over 2 weeks.

5. Stalled Cr → delayed graft function

WHAT YOU SEE

Stalled curve, delayed graft function

WHAT IT ENCODES

Delayed graft function (often dialysis-dependent in the first week) is usually acute tubular necrosis from ischemia-reperfusion, most common in deceased-donor kidneys. It is supported with dialysis and recovers; biopsy excludes rejection if prolonged.

BOARD TRAP

Diagnosing rejection for an oliguric graft in the first days — most early delayed graft function is ATN/ischemia-reperfusion, not rejection.

6. Biopsy = next step (acute injury)

WHAT YOU SEE

Biopsy decision card, ATN vs rejection

WHAT IT ENCODES

When graft function is unexpectedly poor or worsening, allograft biopsy distinguishes ATN from acute rejection and other causes. Doppler ultrasound first rules out obstruction and vascular problems before biopsy. Biopsy guides whether to escalate immunosuppression.

BOARD TRAP

Empirically pulsing steroids for early dysfunction — biopsy (after imaging) separates recoverable ATN from rejection before escalating therapy.

7. Don't over-diurese a slow graft

WHAT YOU SEE

Biopsy card, slow Cr fall don't panic

WHAT IT ENCODES

A slowly recovering graft with low urine output reflects ATN, not necessarily volume overload; aggressive forced diuresis can cause harm. Manage volume carefully and let the tubules recover while monitoring the Cr trend.

BOARD TRAP

Treating low urine output from ATN as simple under-resuscitation and over-diuresing — recovering tubules need time, not aggressive forced diuresis.

8. Day 0: induction + steroids + antimetabolite

WHAT YOU SEE

Day-0 pharmacy cart, triple regimen

WHAT IT ENCODES

Immunosuppression starts on day 0 with induction (often lymphocyte-depleting ATG/alemtuzumab or basiliximab) plus maintenance: a calcineurin inhibitor, an antimetabolite (mycophenolate), and steroids. Induction reduces early acute rejection.

BOARD TRAP

Forgetting induction therapy — day-0 regimens add an induction agent (ATG/basiliximab) on top of CNI + antimetabolite + steroids, not maintenance alone.

9. CNI can be delayed (ATG/alemtuzumab)

WHAT YOU SEE

Delayed CNI waiting room, ATG/alemtuzumab

WHAT IT ENCODES

When delayed graft function/ATN is present, the nephrotoxic calcineurin inhibitor can be delayed by using lymphocyte-depleting induction (ATG/alemtuzumab) to cover immunologically, sparing the recovering tubules from CNI vasoconstriction.

BOARD TRAP

Loading full-dose tacrolimus into an ATN graft on day 0 — CNI can be delayed under ATG/alemtuzumab cover to avoid worsening tubular injury.

10. Immediate urine = good sign

WHAT YOU SEE

Immediate urine tableau, warning level

WHAT IT ENCODES

Brisk immediate urine output (especially with a living donor) signals good early graft function. Massive post-obstructive or osmotic diuresis can follow; replace fluids and electrolytes aggressively to avoid hypovolemia and electrolyte derangement.

BOARD TRAP

Letting post-transplant polyuria run without replacement — large early diuresis needs aggressive fluid/electrolyte replacement to prevent hypovolemia.

11. Replace post-op diuresis aggressively

WHAT YOU SEE

Replace aggressively, Na/K/water losses

WHAT IT ENCODES

Post-transplant diuresis can produce large urinary losses of sodium, potassium, and free water; match input to output to maintain volume and perfusion of the new graft. Underfilling can precipitate prerenal injury to the fresh kidney.

BOARD TRAP

Under-replacing huge early urine output — inadequate replacement causes hypovolemia and prerenal injury to the newly perfused graft.

12. 48h before surgery: control K

WHAT YOU SEE

48 hrs before op, pre-op timing

WHAT IT ENCODES

Plan dialysis within roughly 48 hours before transplant to ensure acceptable potassium and volume status at induction of anesthesia, minimizing intraoperative arrhythmia risk while avoiding excessive fluid removal that leaves the patient hypovolemic.

BOARD TRAP

Aggressive ultrafiltration right before transplant — over-removing volume leaves the patient hypovolemic and underperfuses the new graft.

13. Ureter → bladder anastomosis

WHAT YOU SEE

Remove ileal? ureter to bladder, stent

WHAT IT ENCODES

The donor ureter is implanted into the recipient bladder (ureteroneocystostomy), often over a temporary stent. Early complications include urine leak and obstruction, evaluated with ultrasound; a stent reduces leak/stricture and is removed weeks later.

BOARD TRAP

Overlooking urinary leak/obstruction as a cause of early graft dysfunction — the ureteral anastomosis can leak or obstruct; imaging is the first step.

14. Boards decision band

WHAT YOU SEE

Boards decision band summary icons

WHAT IT ENCODES

Summary logic: pre-op dialyze K if high; graft to iliac fossa, native kidneys stay; Cr should fall by ~2 weeks — if not, image then biopsy; day 0 starts induction + CNI + antimetabolite + steroids, and CNI can be delayed under ATG/alemtuzumab if ATN.

BOARD TRAP

Memorizing steps without the 'if not, biopsy' branch — the board pivot is recognizing when a stalled Cr needs imaging and biopsy versus expectant ATN recovery.